



ORBITGUARD

Free, explainable, **validated** satellite conjunction screening
for the operators commercial space-traffic services price out

99.73%	99.60%	53	11.6 s
RECALL vs 913,330-EVENT FEDERAL ANSWER KEY	PRECISION	AUTOMATED TESTS (CI)	ISS vs 15,697 OBJECTS, 7 DAYS

OrbitGuard · Technical Report

Python · FastAPI · CesiumJS · Next.js · Groq — \$0 infrastructure, 100% public data

1 · THE PROBLEM

Low-Earth orbit is filling up: more than **11,000 active satellites** share the shells below 2,000 km with tens of thousands of tracked debris objects, and a single collision (Iridium-33/Cosmos-2251, 2009) can add thousands of fragments that stay up for decades. Every operator — including university cubesat teams and Indian smallsat startups like Pixxel and GalaxEye — receives **conjunction warnings**: cryptic Conjunction Data Messages with a roughly 99% false-alarm base rate.

The tools that turn those warnings into decisions are **commercial SSA services costing lakhs of rupees per year** (LeoLabs, COMSPOC) — priced for constellations, not for a team flying one satellite. The free alternatives are tables without explanations (SOCRATES), visualizations without validated math (most globe demos), or raw catalogs. None of them answer the operator's actual question:

THE OPERATOR'S QUESTION

"Do I need to move my satellite, and can I trust the number that says so?"

2 · THE SOLUTION

OrbitGuard is a browser-based conjunction-assessment system built entirely on free, public data. It screens an operator's satellites against the full public catalog, computes collision probability with the **same class of method NASA CARA runs operationally**, triages every event into an explained **DODGE / WAIT / WATCH** verdict, and narrates the evidence in plain language. Its defining property:

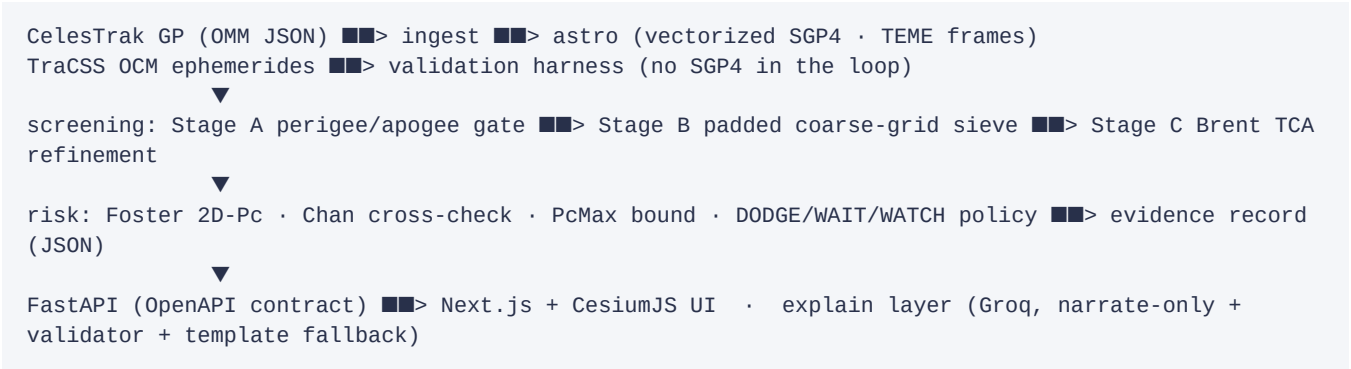
THE HONESTY PRINCIPLE

Public TLE data carries no error model, so OrbitGuard never invents one. It computes the **worst-case** collision probability over every error model consistent with the geometry: $P_{cMax} = R^2/(e \cdot d^2)$. An upper bound can **prove an event safe** (P_{cMax} below threshold under any covariance) but can **never order a maneuver** — for that, the verdict escalates to requesting CDM-grade data. TLE data can prove safety, never danger.

Key features

- **Mission Control:** live 3D globe (CesiumJS) with animated assets, the ambient catalog, ranked verdict cards with countdown-to-TCA, and the screening funnel with timings.
- **Encounter Inspector:** per-event flyby view, encounter-plane figure (hard-body circle, miss vector, worst-case error circles), full numeric evidence record with raw-JSON toggle.
- **Validation Scorecard:** recall/precision against the official U.S. TraCSS answer key, rendered in-app with dataset citation and git SHA — we do not mark our own homework.
- **Grounded AI explanations:** a narrate-only LLM (Groq, llama-3.3-70b) converts the final evidence JSON to prose; a validator rejects any output containing numbers not present in the evidence; a deterministic template renderer is the always-on fallback.
- **India story:** demo assets are ISS, Pixxel FIREFLY-1, GalaxEye DRISHTI (NORAD 69010 — adjacent to the July-2026 5-digit catalog rollover our OMM-first design survives), ISRO EOS-08.

3 · ARCHITECTURE



Data-flow rule that keeps the system honest: every number shown anywhere is computed in the Python data plane. The LLM converts one finished JSON record into sentences; it cannot change a verdict, a probability, or a distance. OrbitGuard is not an AI wrapper — it is a numerical pipeline with an explanation layer bolted on the end.

Layer	Choice	Why
Propagation	python-sgp4 (vectorized SatrecArray) + Skyfield frames	GP elements are fitted to SGP4 — anything else de-calibrates them; TEME→GCRS cross-checked < 5 m vs Skyfield
Screening	3-stage sieve, numpy + cKDTree	provable no-miss pad; 15,697 objects → 7 events in 11.6 s
Probability	Foster-1992 quadrature + Chan-1997 series + PcMax bound	NASA CARA's operational method; two independent implementations cross-check in CI
API	FastAPI + pydantic	frozen OpenAPI contract decouples frontend
UI	Next.js 14 + CesiumJS + satellite.js	offline NaturalEarthII imagery — no ion token, demo survives dead Wi-Fi
LLM	Groq free tier, llama-3.3-70b-versatile	narrate-only contract, digit validator, deterministic fallback
Hosting	\$0 — laptop / free tiers	committed fixtures make offline demo first-class

4 · THE ALGORITHMS (AND WHY THEY ARE RIGHT)

4.1 The screening sieve. Stage A discards objects whose perigee/apogee bands cannot meet the asset's (Hoots Filter I). Stage B propagates survivors on a coarse grid ($\Delta t = 30$ s live, 8 s all-vs-all) and flags any pair within the padded radius:

$$R_{\text{pair}} = D + v_{\text{rel,max}} \cdot \Delta t / 2 \quad (v_{\text{rel,max}} = 15.6 \text{ km/s, head-on LEO worst case})$$

If a conjunction with miss $< D$ occurs at t^* , the nearest grid point lies within $\Delta t/2$ and the pair can separate by at most $v_{\text{rel,max}} \cdot \Delta t/2$ — so its grid separation is below R_{pair} . **No true conjunction can escape the coarse net** (Alarcón-Rodríguez et al., ESA 2002). Stage C finds the exact TCA as a root of $g(t) = \Delta r \cdot \Delta v$ with Brent's method.

4.2 Collision probability. At LEO closing speeds the encounter is instantaneous and the 3D problem collapses onto the 2D plane perpendicular to the relative velocity. The probability is a 2D Gaussian integrated over the combined hard-body disk:

$$P_c = (2\pi)^{-1} |C_p|^{-1/2} \iint_{|x| \leq R} \exp(-(x-m)^T C_p^{-1} (x-m)/2) dA \quad [\text{Foster 1992 — NASA CARA's operational method}]$$

Foster (adaptive quadrature) is the displayed value; Chan's Rician series is computed alongside as a cross-check — an axis swap or frame bug moves P_c by orders of magnitude and trips the CI alarm. When no covariance exists (all public TLE data), the worst-case bound $P_{c\text{Max}} = R^2/(e \cdot d^2)$ is reported instead, with the policy asymmetry of Section 2.

Data grade	Condition	Verdict
CDM-grade	$P_c \geq 10\blacksquare\blacksquare\blacksquare$	DODGE — NASA maneuver-consideration threshold
CDM-grade	$10\blacksquare\blacksquare \leq P_c < 10\blacksquare\blacksquare\blacksquare$	WATCH — monitoring band
CDM-grade	$P_c < 10\blacksquare\blacksquare$	WAIT
TLE-grade	$P_{c\text{Max}} < 10\blacksquare\blacksquare$	WAIT — provable clearance under any covariance
TLE-grade	$P_{c\text{Max}} \geq 10\blacksquare\blacksquare$ and miss < 5 km	WATCH + ESCALATE : request CDM / owner ephemeris
any	$v\blacksquare\blacksquare\blacksquare < 100$ m/s	WATCH + flag: 2D- P_c not applicable (disclosed, not faked)

5 · VERIFICATION — MEASURED, NOT ASSERTED

Ring 1 — continuous (53 tests, every commit).

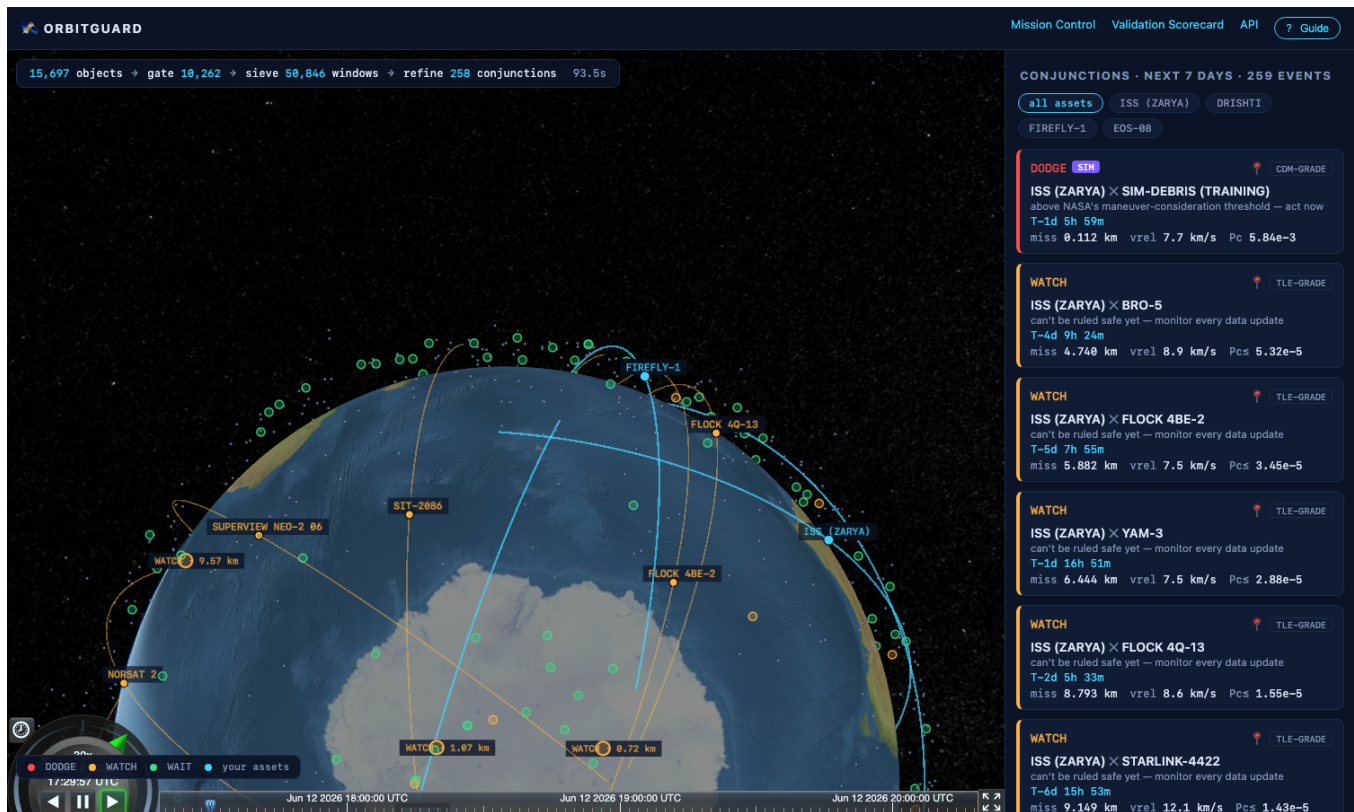
Claim	Independent referee
The sieve misses nothing	brute-force 1 s search over every pair — recall AND precision asserted on scenes with planted conjunctions, near-misses, GEO/Molniya decoys
Foster Pc is exact	closed-form noncentral- χ^2 (isotropic), scipy dblquad referee (anisotropic, machine precision), 2M-sample Monte Carlo
PcMax is a theorem	σ -sweep over 4 decades: exact Pc never exceeds $R^2/(e \cdot d^2)$, bound attained at $\sigma = d/\sqrt{2}$
Frames are right	Skyfield's independent TEME→GCRS pipeline, < 5 m
Harness scores correctly	synthetic ephemerides with closed-form TCA/miss; duplicate events penalize precision

Ring 2 — the federal exam. The U.S. Office of Space Commerce (TraCSS program) publishes a *Dataset for Conjunction Assessment Verification* (The Aerospace Corporation, 2026, CC0): 20.7 GB of CCSDS-OCM ephemerides — the real TLE-derived catalog plus synthetic maneuvering objects, historical-CDM reconstructions and the OSIRIS-REx reentry — with a **913,330-conjunction answer key** generated by Aerospace's validated CSieve tool. OrbitGuard's screening engine runs directly on those ephemerides (no SGP4 in the loop — the key tests screening, not propagation), all-vs-all over 26,793 ephemerides, 7-day window, on an 8 GB laptop.

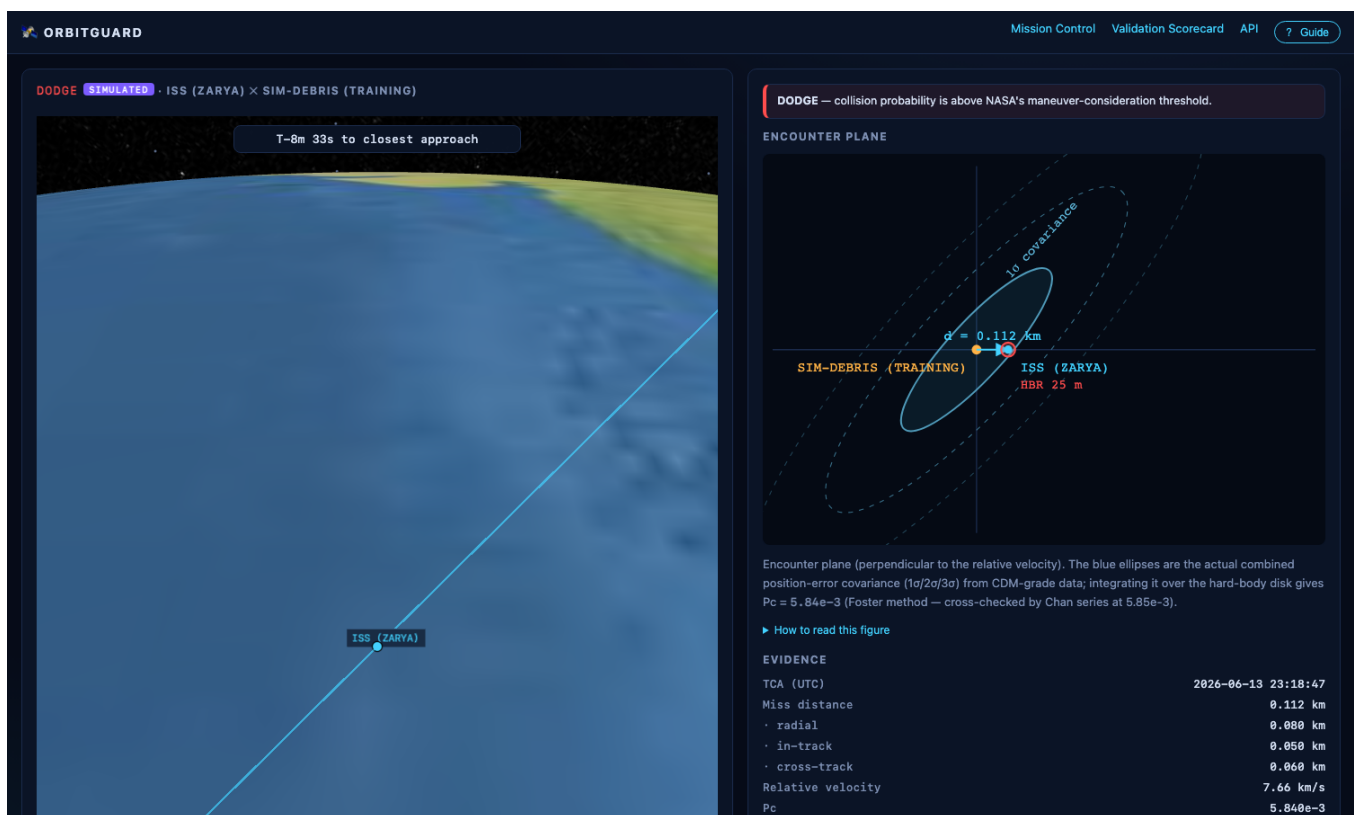
99.73%	99.60%	61.34 ms	0.14 m
RECALL	PRECISION	TCA RMS ERROR	MISS RMS ERROR

Matching: unordered pair + $|\Delta TCA| \leq 5$ s, each key row claimable once. Run configuration per the official User's Guide: spherical 10 km volume, usable-window bounds honored, OD epoch < 14 days, TCA strictly inside 2025-01-01T12:00→2025-01-08T12:00 UTC. Disclaimer carried verbatim in-app: the dataset is for algorithm testing, not operational certification.

6 · THE PRODUCT



Mission Control — live globe, animated assets and top-risk markers, ranked verdict cards with countdowns, and the screening funnel strip (15,697 objects → events, with timings).



Encounter Inspector — encounter-plane figure (hard-body circle, miss vector, worst-case σ circles), full evidence grid, grounded LLM explanation with raw-JSON toggle.

Validation Scorecard

OrbitGuard's screening engine is graded against the official U.S. TraCSS *Dataset for Conjunction Assessment Verification* — 20.7 GB of federal test ephemerides with a 913,330-conjunction answer key generated by The Aerospace Corporation's validated CSieve tool (CC0).

FULL 7-DAY RUN

• 2025-01-01T12:00:00+00:00 +168h • 26,793 ephemerides, all-vs-all • engine runtime 220 min

@3f66f15

99.73% RECALL vs ANSWER KEY	99.60% PRECISION	61 TCA RMS ERROR (ms)	0.1 MISS RMS ERROR (m)
910,841 EVENTS MATCHED	914,495 EVENTS FOUND	913,330 ANSWER-KEY EVENTS	99.66% F1

Methodology

1. Stage B+C screening run directly on the provided OCM ephemerides (no SGP4 in the loop)
2. 6-point Lagrange interpolation on position, analytic velocity derivative
3. All-vs-all, usable windows honored, OD epoch < 14 d, same-designator pairs excluded
4. Matched to the spherical DefaultHBR key on (unordered pair, |dTCA| <= 5 s)
5. Duplicates count against precision; each key row claimable once

Honest limits. TLE/GP accuracy is ~1 km at epoch and degrades 1–3 km/day (in-track dominant). Without covariance, OrbitGuard reports a worst-case Pc bound: it can prove an event is safe under any consistent error model, but it can never prove danger — that requires CDM-grade data. Low relative-velocity encounters (< 100 m/s) violate the 2D-Pc assumptions and are flagged, not faked. The answer key itself is documented as "expected to find nearly all events" rather than exhaustive — residual disagreements are published in the autopsy file, not hidden.

Dataset disclaimer: "Dataset is for algorithm testing, not operational certification (per the Office of Space Commerce)."

Validation Scorecard — the federal-answer-key metrics rendered in-app, with methodology and honest-limits statement.

7 · DATA SOURCES (ALL FREE, ALL PUBLIC)

Source	What	Discipline
CelesTrak GP	full active catalog as OMM JSON (15,697 objects)	2 h refresh cadence enforced in code; every snapshot cached; OMM-first — integer catalog IDs survive the July-2026 5-digit rollover
TraCSS verification dataset	20.7 GB OCM ephemerides + CSieve answer keys (CC0)	downloaded once to external SSD; .npy cache; subset-first runs
Space-Track CDMs	covariance for rigorous Pc (stretch)	nightly batch, throttled
Groq free tier	llama-3.3-70b narration	30 RPM / 100K tokens-day budget; SQLite cache; template fallback

8 · HONEST LIMITS

- GP/TLE accuracy is ~1 km at epoch, degrading 1–3 km/day (in-track dominant) — which is exactly why TLE-grade verdicts are worst-case bounds with an escalation path, never maneuver orders.
- Low relative-velocity encounters (< 100 m/s) violate the short-encounter 2D-Pc assumptions; OrbitGuard flags them instead of printing a wrong number.
- Chan's series is an approximation: ~2% near the operational band, tens of percent in the deep tail — measured against an independent integration referee and documented; Foster is the displayed reference everywhere.
- OrbitGuard is a research-grade system, not an operational CA service; the TraCSS dataset is a diagnostic, not a certification.

9 · WHAT THIS PROJECT DEMONSTRATES

A production-shaped system on real physics: vectorized SGP4 screening with a provable completeness guarantee; NASA-method collision probability verified against exact closed forms, independent integrators and Monte Carlo; a risk policy derived from an operational handbook with a defensible information-theoretic asymmetry; a **913,330-event federal answer-key validation** executed end-to-end on consumer hardware via deliberate vectorization (40× refinement speedup); a strictly grounded LLM layer; CI, typed contracts, offline-first demo engineering — built in days, documented, tested, and reproducible.

References: Foster & Estes 1992 (NASA JSC-25898) · Chan 1997/2008 · NASA CARA 2D-Pc recommendations (NTRS 20190028900) · NASA CA Handbook v2 (NTRS 20230002470) · Alarcón-Rodríguez et al., ESA SP-486 (2002) · Hoots et al. 1984 · Vallado et al. 2006 · Auman et al., AMOS 2025 (TraCSS validation methodology) · Office of Space Commerce, Dataset for Conjunction Assessment Verification (CC0) · CelesTrak GP documentation. Full list in docs/ALGORITHMMS.md.